

PANKAJ THAKUR

Software Engineering | Backend Development | AI / Generative AI

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PROFESSIONAL SUMMARY

AI & Data Science undergraduate with proven hands-on experience in **Full Stack Development**, **Backend Engineering**, and **Generative AI** application development. Engineered production-ready REST APIs, full-stack web systems, and LLM-powered RAG pipelines using React.js, Node.js, Express.js, FastAPI, MongoDB, and LangChain. **GATE CSE 2026 Qualified** — demonstrating strong algorithmic thinking, problem-solving ability, and core CS fundamentals.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, HTML, CSS

Frontend: React.js, Tailwind CSS

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication

Databases: MongoDB, MySQL

AI / GenAI: LangChain, RAG, ChromaDB, HuggingFace Embeddings, NLP, LLM Applications

Core CS: Data Structures & Algorithms, DBMS, OOP, Operating Systems, Computer Networks

Tools: Git, GitHub, Postman, VS Code

PROJECTS

SmartPark — Intelligent Parking Management System

[GitHub](#) | [Live Demo](#)

Full Stack Developer / Backend Engineer

- Architected a full-stack parking management platform featuring real-time slot availability, automated booking flows, and integrated payment processing with role-based access for admin and user roles.
- Engineered a secure REST API backend with JWT-based authentication and RBAC, enabling distinct permission models across admin and user-facing dashboards.
- Designed an optimized MongoDB schema for high-throughput reservation tracking and built a responsive React.js frontend with clean UI for seamless UX.

Stack: React.js · Node.js · Express.js · MongoDB · REST APIs · JWT Authentication

DocMind AI — Intelligent PDF Question-Answering Chatbot

[GitHub](#)

AI Application Developer / Backend Engineer

- Built an AI-powered document chatbot using Retrieval-Augmented Generation (RAG) architecture, enabling accurate context-aware Q&A across multi-document PDF corpora.
- Engineered a full ingestion pipeline: PDF parsing → semantic chunking → HuggingFace embedding generation → ChromaDB vector storage → similarity-based retrieval.
- Integrated Mistral AI LLM via LangChain for multi-turn conversational querying with contextual memory, deployed via an interactive Streamlit interface.

Stack: Python · LangChain · ChromaDB · Mistral AI · HuggingFace Embeddings · Streamlit · RAG

EDUCATION

Madhav Institute of Technology & Science, Gwalior

2023 – 2027

B.Tech — Artificial Intelligence & Data Science CGPA: **8.3 / 10**

Coursework: Data Structures & Algorithms · DBMS · OOP · Operating Systems · Machine Learning · NLP · Computer Networks

ACHIEVEMENTS

GATE CSE 2026 Qualified — AIR **14,658** | Score: **426** | Competed among ~2 lakh+ candidates nationwide, validating strong CS fundamentals and analytical ability.

[LeetCode](#) — **Problem Solving** | Solved **120+ problems** in C++ across Arrays, Linked Lists, Trees, Graphs, Hashing, Dynamic Programming, Sorting & Searching.

CERTIFICATIONS

Data Structures & Algorithms — [PW Skills](#)

Backend Development — [Udemy](#)